A3-PS010-TC099

Evidence-Based IP Network-Type Classification

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Problem Description

- Fraud engines must tell residential, data centre, VPN and Tor traffic apart.
- Static databases misclassify legitimate users.
- Hosting alone is not proof of VPN use.
- Tunnel services such as Apple iCloud Private Relay are often missed.
- Insufficient evidence must return Unknown.

Background

One public IP can serve many users (RFC 6269), so an IP alone is weak evidence.

Tor exit list	VPN and relay ranges	Network context
Onionoo (Tor Project)	X4BNet list, Apple egress	ASN and IP-Atlas

Literature & RFC Review

RFC 6269: describes shared IP addresses through carrier-grade NAT.

Tor Project Onionoo: official public list of running Tor exit relays.

Apple iCloud Private Relay: Apple's published egress ranges for its relay service.

Proposed Solution

An evidence-based classifier with an audit trail.

Known Tor Exit

Known Hosted VPN/Proxy Endpoint

Known Tunnel/Relay Endpoint

Unknown

shown as HOSTING ONLY or NO EVIDENCE

Every result shows its rule, source, live or snapshot status, and fetch time.

Optimization Proposed

Area	Before	After
Tunnel detection	Apple relay IPs labelled VPN	Separate Tunnel/Relay label
Rule order	VPN list checked first	Tunnel ranges checked first
Unknown display	Amber warning	Neutral grey: NO EVIDENCE or HOSTING ONLY
Range safety	All ranges used	Malformed, too broad or private ranges rejected
Lookup	Scan every range	Indexed lookup: 270,000 ranges in about 3.5 s

Solution Architecture & Design

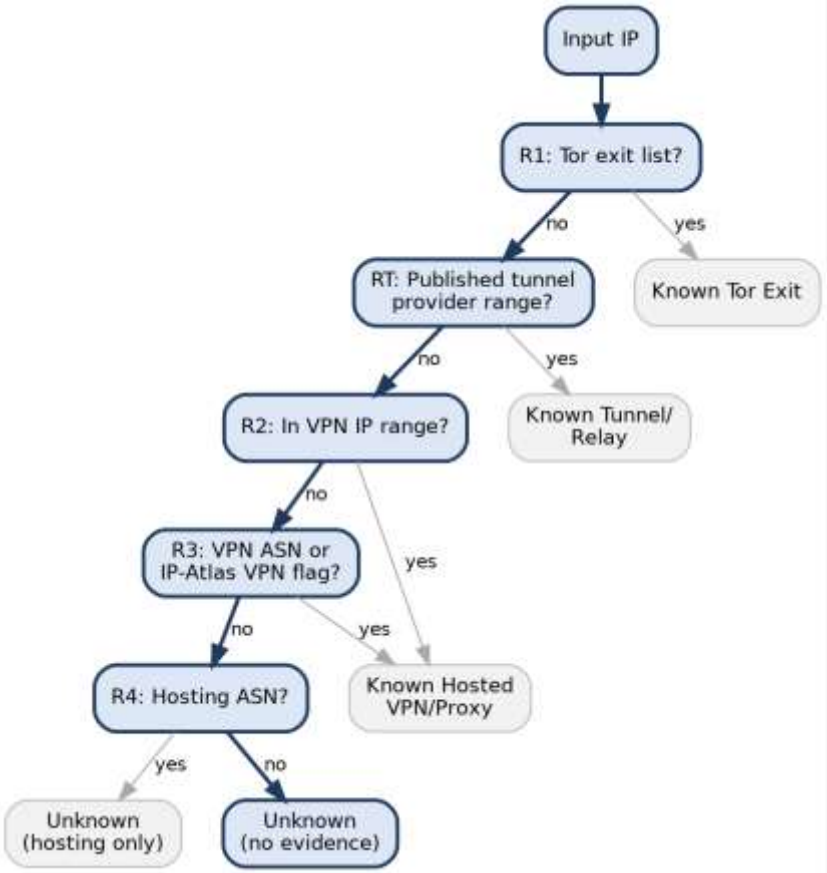


Figure: Decision flow. Rules run in order and stop at the first match.

Data Flow Diagram

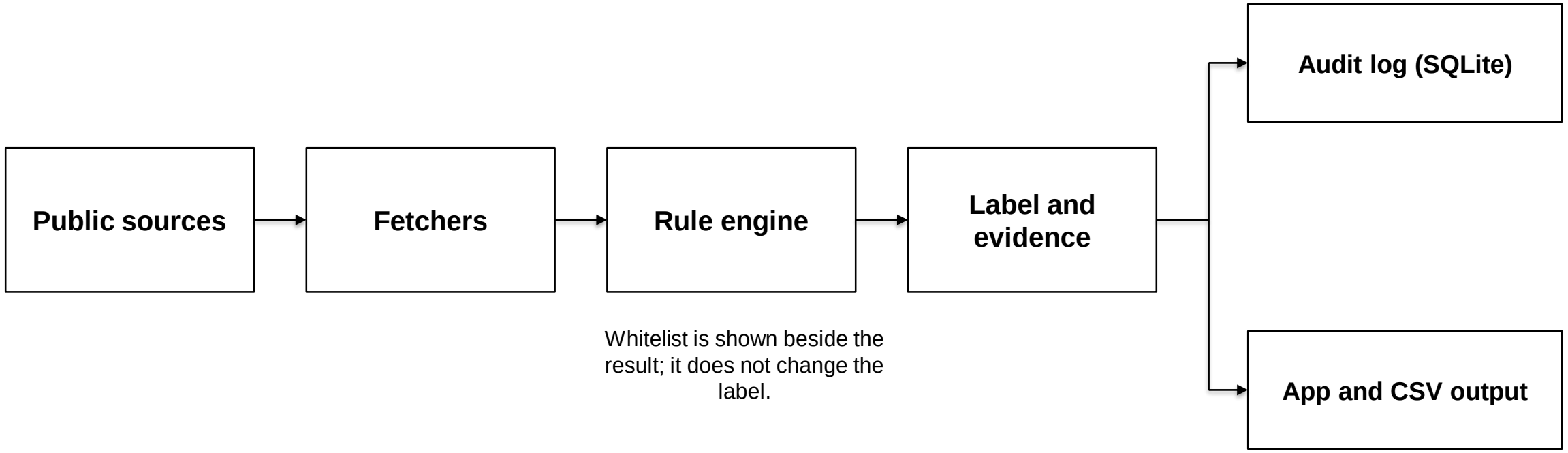
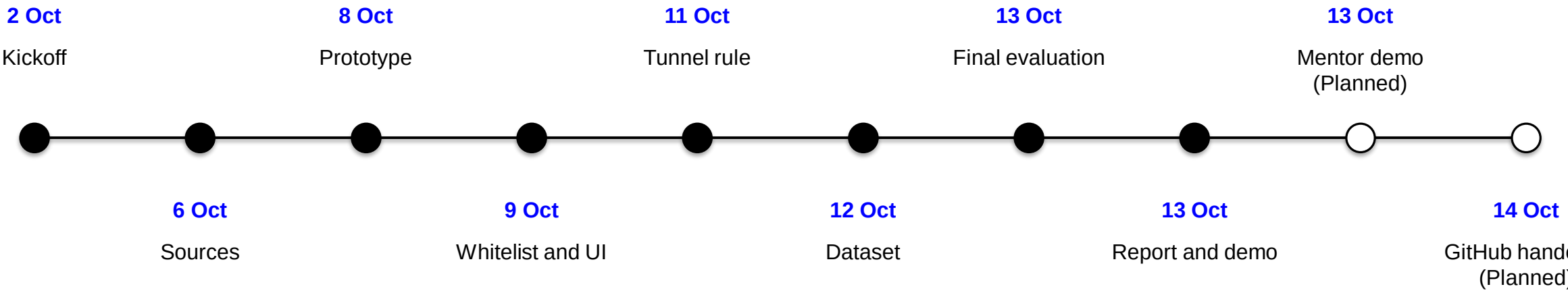


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Eight milestones done; two planned.



Bibliography

1. RFC 6269, Issues with IP Address Sharing
2. Tor Project, Onionoo protocol
3. X4BNet, VPN IP range list
4. Apple, iCloud Private Relay egress ranges
5. NordVPN and Mullvad server APIs
6. AWS published IP ranges
7. IPinfo Lite API and IP-Atlas API
8. scikit-learn and Streamlit

Thank You for Your Attention!
TEAM ARGUS